

3a	<u>Resultant/net force</u> (in any direction) on the object must be <u>zero</u> and <u>Resultant/net moment / torque</u> on the object <u>about any point / axis</u> must be zero.	B1 B1
3b	Taking moments about end A, $(W \times 0.25) + (12 \times 0.35) = (17 \sin 50^\circ \times 0.50)$ $W = 9.246$ $= 9.2 \text{ N}$	C1 A1
3c	Consider vertical equilibrium, taking upwards as positive Sum of forces in vertical direction = 0 $F_y + 17 \sin 50^\circ - 9.2 - 12 = 0$ $F_y = 8.177 \text{ N}$ Consider horizontal equilibrium, taking rightwards as positive Sum of forces in horizontal direction = 0 $17 \cos 50^\circ - F_x = 0$ $F_x = 10.93 \text{ N}$ $F = \sqrt{(8.177^2 + 10.93^2)} = 13.7 \text{ N}$	C1 C1 A1
3d	By taking moments about end A, The <u>moment due to the force by the block on the beam decreases</u> , , <u>the tension in the string decreases</u> .	M1 A1

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	When the tension in the string decreases at the same angle, by considering <u>horizontal equilibrium</u> , the horizontal component of the force exerted on the beam by the hinge <u>decreases</u> .	
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4ai	reduction in energy / amplitude (of the oscillations)	B1
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